

CogniFlow AI - Project Report

Multimodal Intelligence Platform for Emotion Analysis, RAG, OCR, and Movie Recommendation

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Executive Summary

CogniFlow AI is an integrated platform that combines document understanding, OCR-based image intelligence, multimodal emotion analysis, and recommendation workflows in a single production-oriented interface. The project was designed to reduce fragmentation between separate AI tools by providing one unified system for analysis and decision support. The backend provides the core AI pipelines and APIs, while the frontend delivers a modern dashboard for interaction, visualization, and reporting.

1. Background

AI systems for text, image, and video often exist as isolated modules with separate user experiences and inconsistent outputs. In practical environments such as media intelligence, educational content analysis, and enterprise knowledge workflows, this creates operational friction, low explainability, and poor usability. CogniFlow AI addresses this by unifying multimodal capabilities under one architecture and one interface.

2. Problem Statement

The core problem is lack of an integrated, explainable, and scalable platform for multimodal intelligence. Existing solutions typically suffer from one or more of the following: black-box recommendation logic, weak traceability, non-uniform API contracts, poor frontend visualization for analysis outputs, and limited flexibility between local and API-driven model execution.

3. Project Objectives

- Build a unified AI platform for video, script, PDF, image, and chat workflows.
- Provide accurate emotion analysis with timeline outputs and meaningful visualizations.
- Integrate OCR and document retrieval for image/document question answering.
- Deliver deterministic and explainable movie recommendations from structured scoring.
- Support robust frontend UX with production-grade navigation, loading states, and reporting.

4. Methodology

4.1 System Architecture

The system uses a modular architecture with clearly separated layers: input ingestion, preprocessing, model inference, recommendation logic, persistence, and frontend presentation. Backend services are exposed through FastAPI endpoints. The frontend is implemented in React + TypeScript and communicates via a centralized API service layer.

4.2 Backend Components

Component	Purpose
Script Analyzer	Emotion classification with sentence-level timeline
Video Pipeline	Transcription + emotion/intensity processing
PDF Pipeline	Text extraction + script-style emotion analysis
Image OCR Pipeline	Text extraction from images and solve/index modes
Recommendation Engine	Deterministic movie ranking by filter/mood matching
RAG Components	Document indexing and context retrieval for QA

4.3 Frontend Method

Frontend design followed a reusable component strategy with motion-enabled transitions, dashboard shortcuts, and module-specific pages (Video, Script, PDF, Image Search, Chatbot, Recommendations). API response normalization was introduced to handle differences in backend payload shape (for example emotional arc fields), ensuring stable rendering.

5. Results and Outcomes

- All core analysis modules were integrated into a single UI with unified navigation.
- Script/PDF output rendering issues were resolved by robust response normalization.
- Emotion detection stability was improved by weighted score aggregation in the backend analyzer.
- Recommendation quality was improved via case-insensitive matching, mood alias expansion, and better fallback behavior.
- Global UX improvements added: animated transitions, branded loader, dashboard shortcuts, and clearer chatbot message formatting.

6. Validation and Testing

Validation included backend syntax checks, endpoint contract alignment, and frontend build verification. Project-level checks were run after major fixes to prevent regressions. The UI build compiles successfully and major analysis flows now return visible, consistent outputs.

7. Conclusion

CogniFlow AI successfully demonstrates a practical multimodal AI platform with unified workflows across text, document, image, and video intelligence. The project now has stronger consistency between backend outputs and frontend rendering, improved recommendation relevance, and a more maintainable architecture for future extension.

8. Supporting Documentation

- Backend framework: FastAPI
- Frontend stack: React, TypeScript, Tailwind CSS, Framer Motion
- Database and ORM: SQLite, SQLAlchemy
- NLP/ML toolchain: HuggingFace Transformers, sentence embeddings, vector retrieval
- Key modules documented in codebase: `app/main.py`, `app/video/script_analyzer.py`, `app/recommendation/engine.py`, `ui/src/services/api.ts`

9. Future Improvements

- Add benchmark-driven evaluation dashboard (precision/recall/confusion trends).
- Introduce multilingual emotion classification and translation fallback.
- Add richer report generation templates with charts embedded in exported PDFs.
- Add user-level analytics and role-based access for enterprise deployment.